MULTIPLE STRESSORS IN THE FUTURE WESTERN BALTIC SEA: A CLIMATE-DRIVEN ECOSYSTEM MODEL

Marcela Nascimento & Marco Scotti

Ecosystem Modelling Working Group

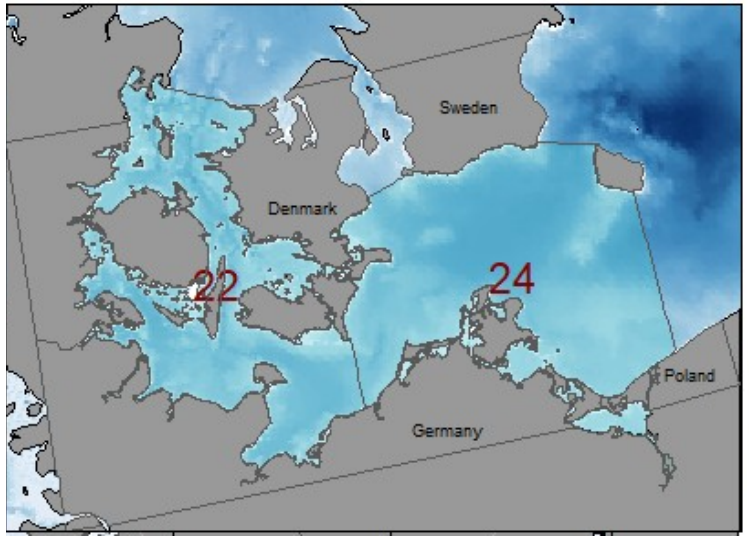
Research Division: Marine Ecology

Research Unit: Marine Evolutionary Ecology

Main question

Will plausible projected global warming and fisheries scenarios impact the functional groups' future distribution in the Western Baltic Sea (WBS)?

ICES subdivision 22 and 24

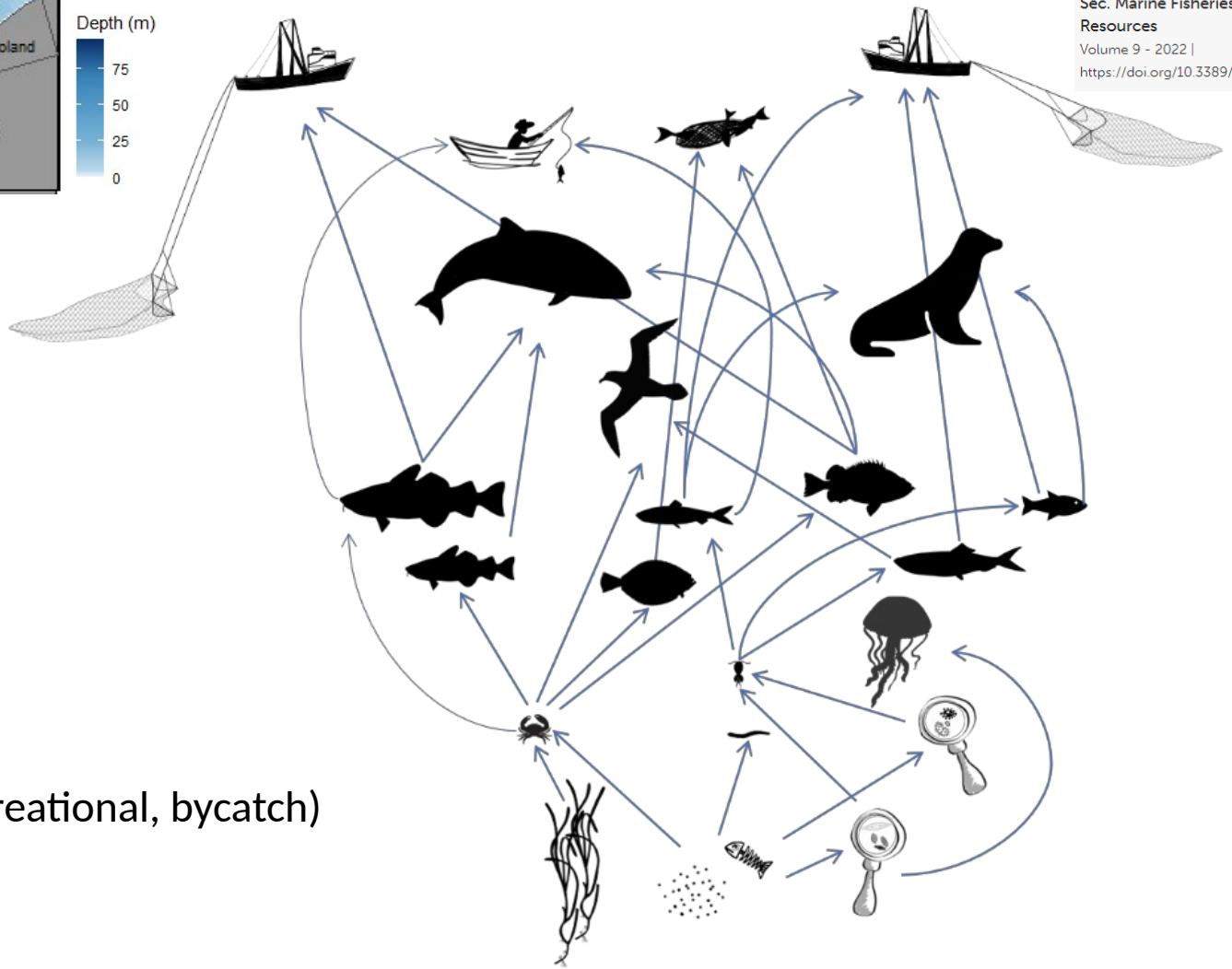


frontiers | Frontiers in Marine Science

Ecosystem-based fisheries management increases catch and carbon sequestration through recovery of exploited stocks: The western Baltic Sea case study

Marco Scotti[1,2*], Silvia Opitz[1], Liam MacNeil[1], Axel Kreutle[3], Christian Pusch[3] and Rainer Froese[1]

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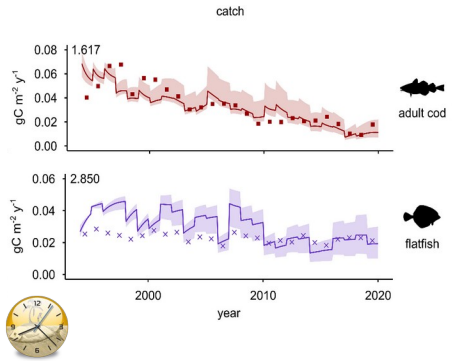
18 Functional groups

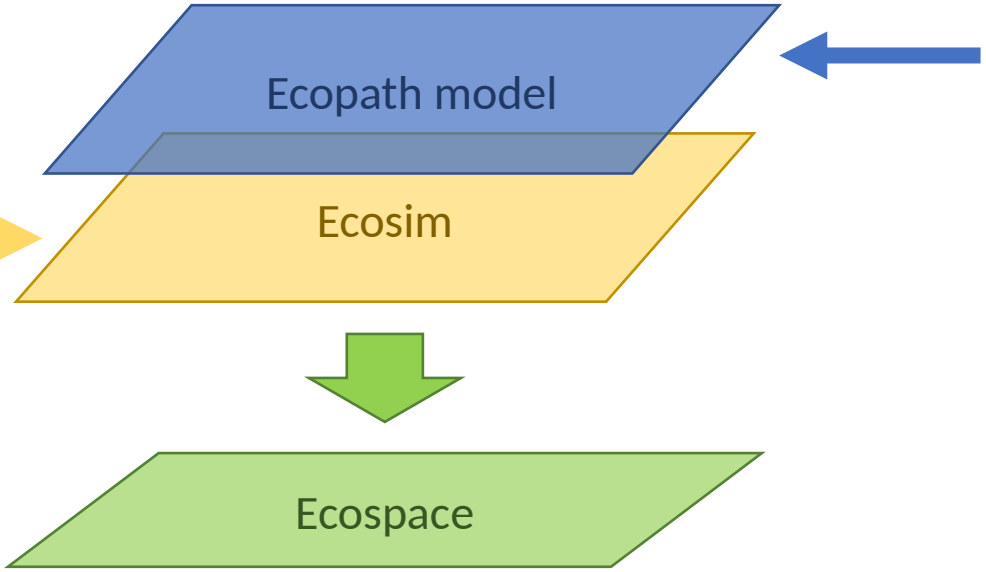
4 Fleets (pelagic, demersal, recreational, bycatch)

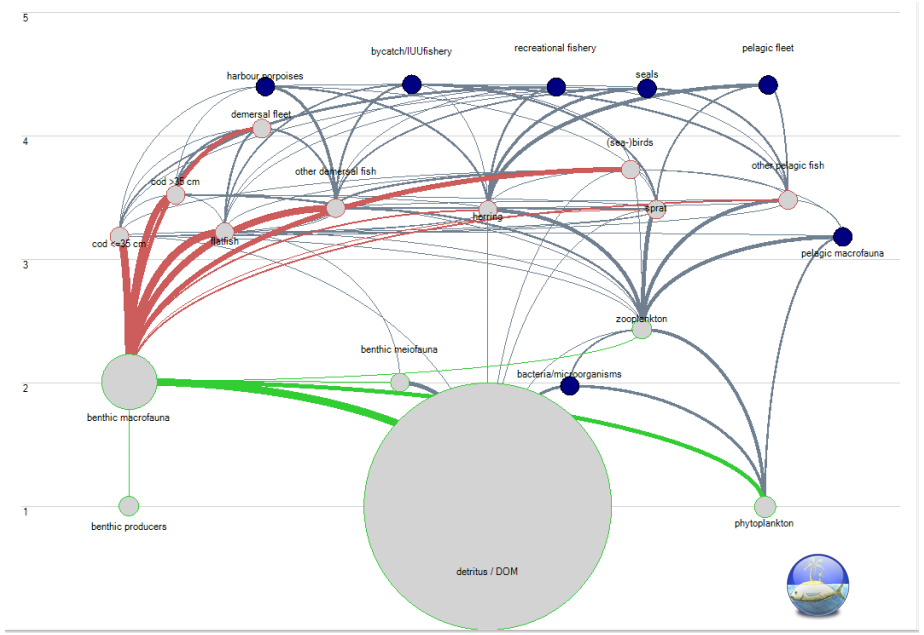
A Snapshot of the Year 1994



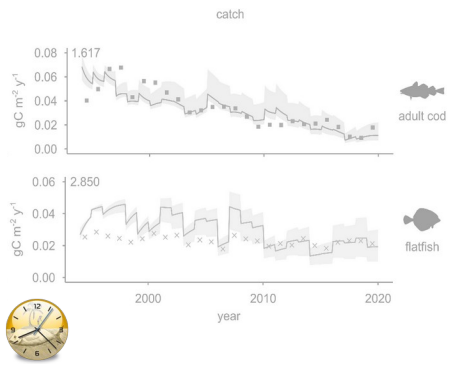
How does Ecospace work?

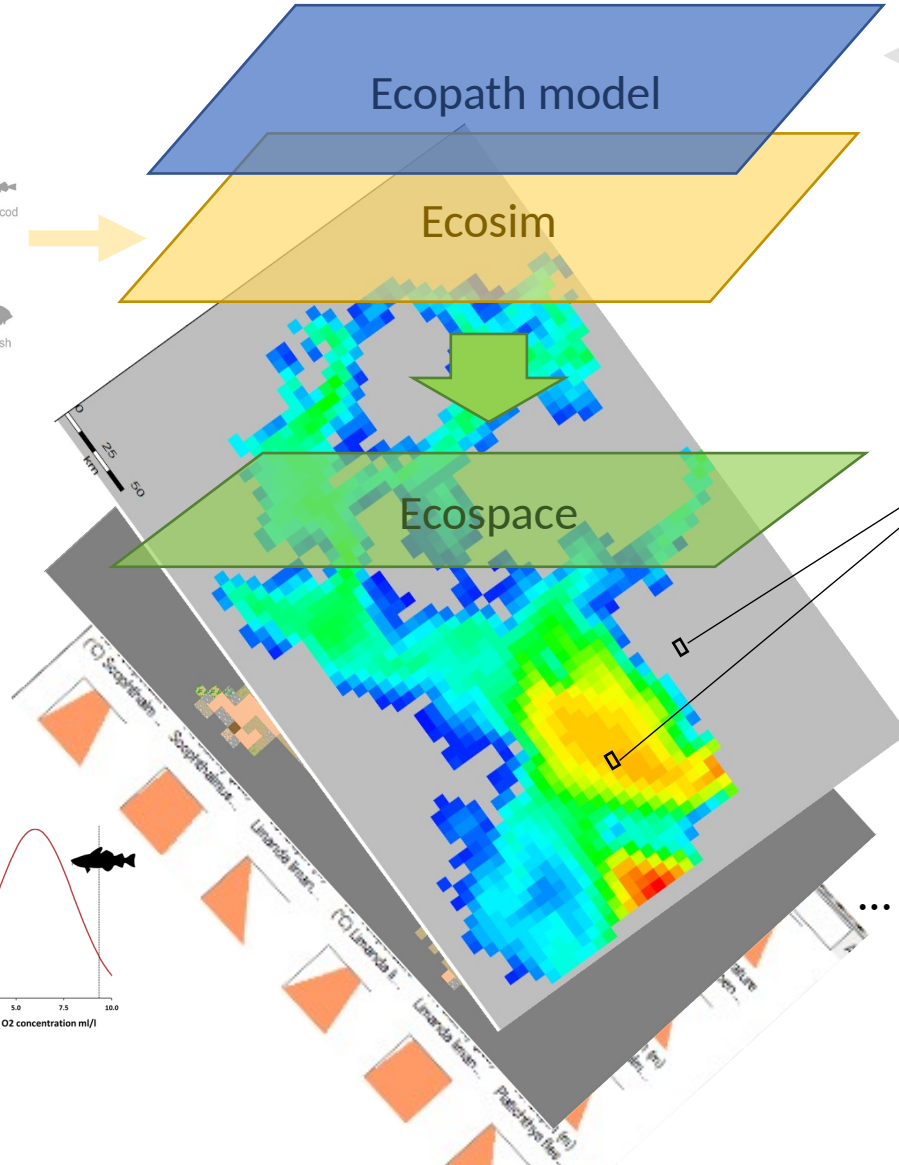


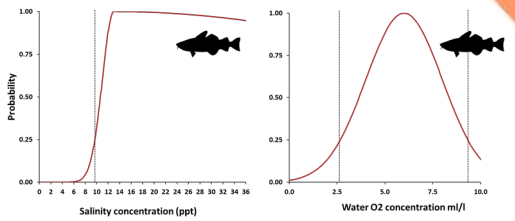


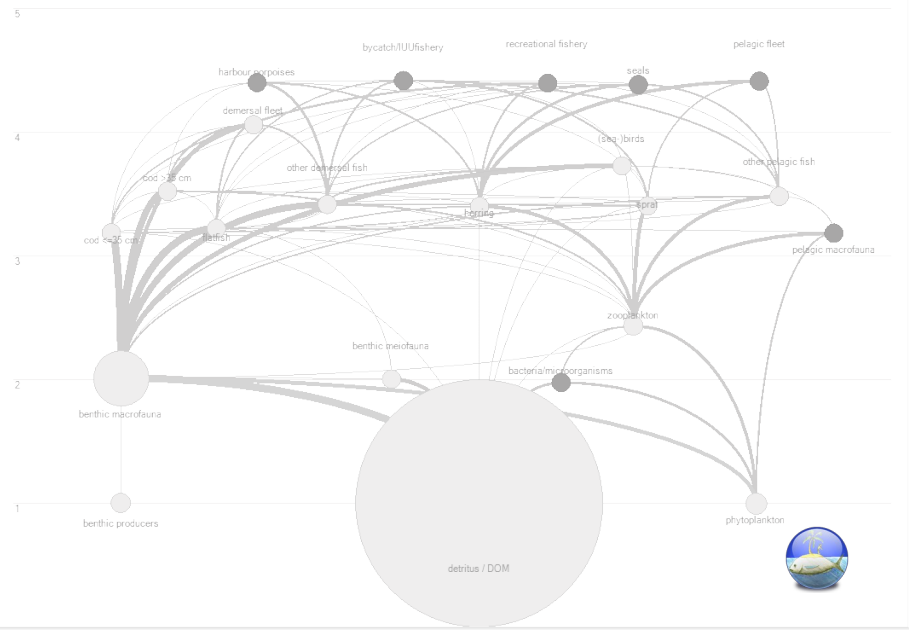


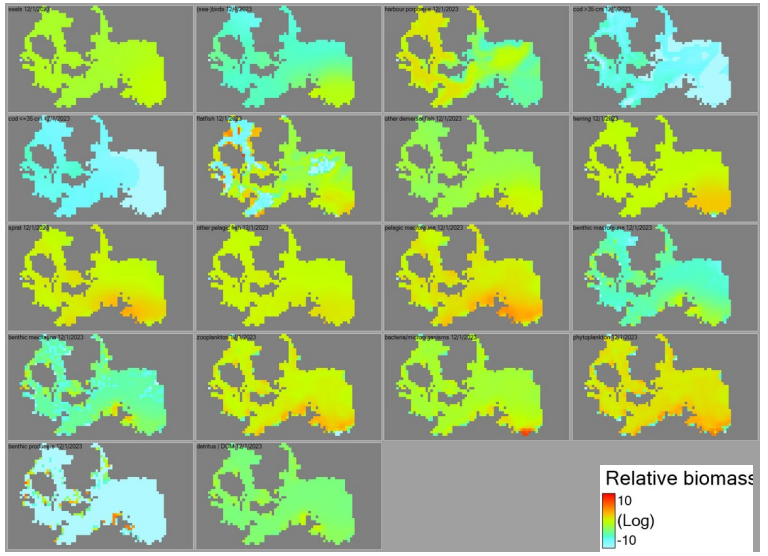
How does Ecospace work?







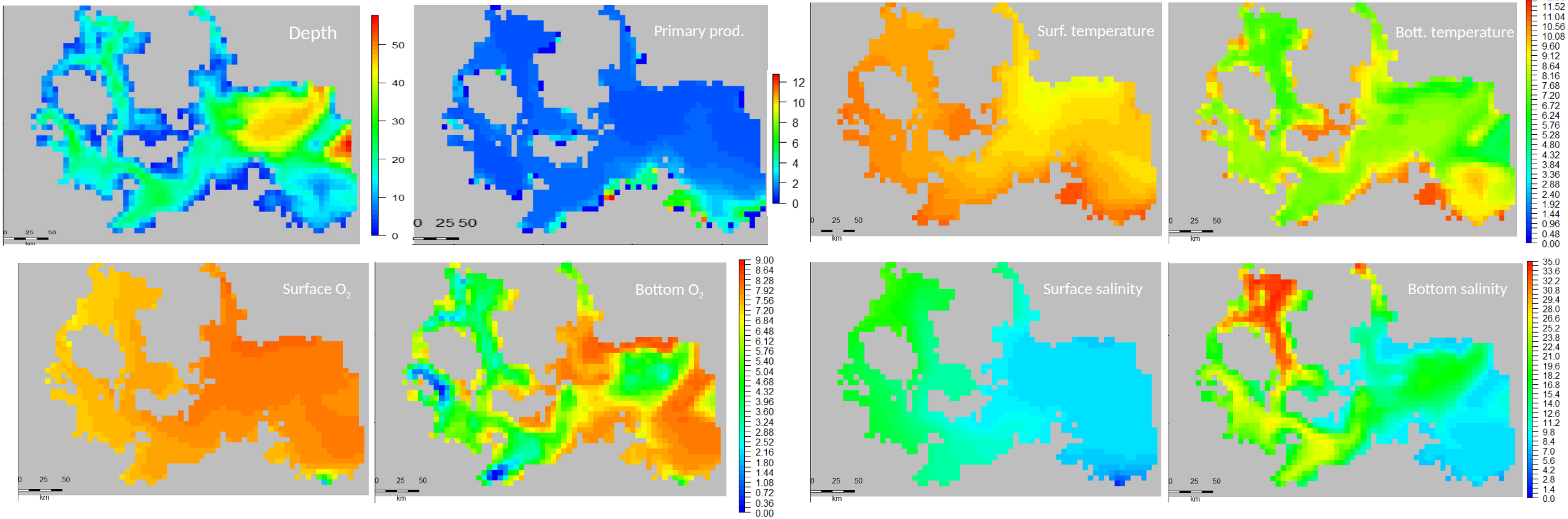




Environmental layers

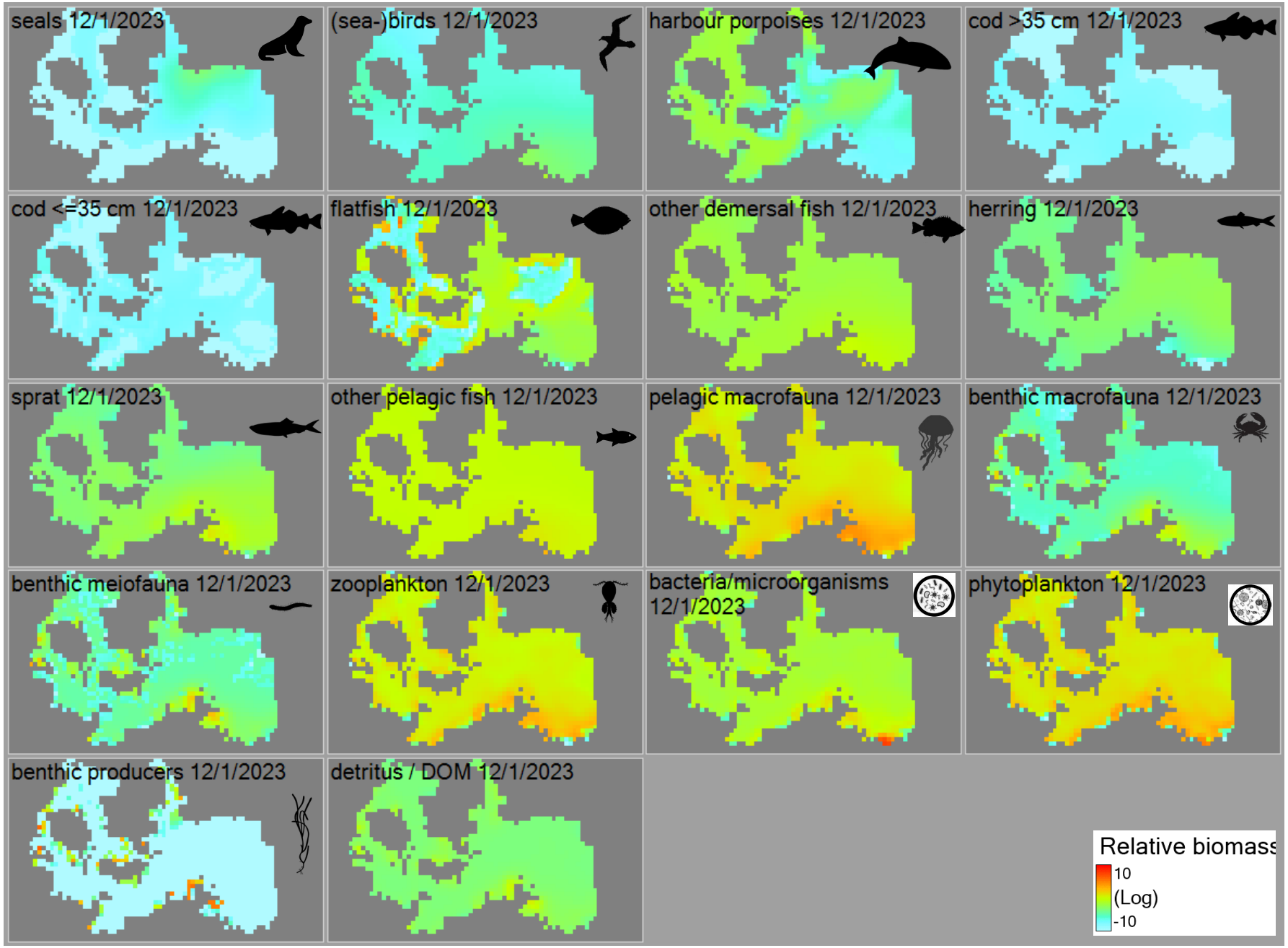
Simulations 1994 – 2100
Monthly imputs

- **Scenarios**
  - **Catch –** BAU, EBFM, F_{MSY}, no fishing
  - **Climate –** SSP - RCP 4.5, 8.5



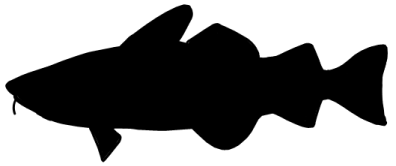
Average of MPI-ESM-LR, EC-EARTH, IPSL-CM5A and HadGEM2 models, all under the RCO-SCOBI models

Preliminary distributions – Relative biomass

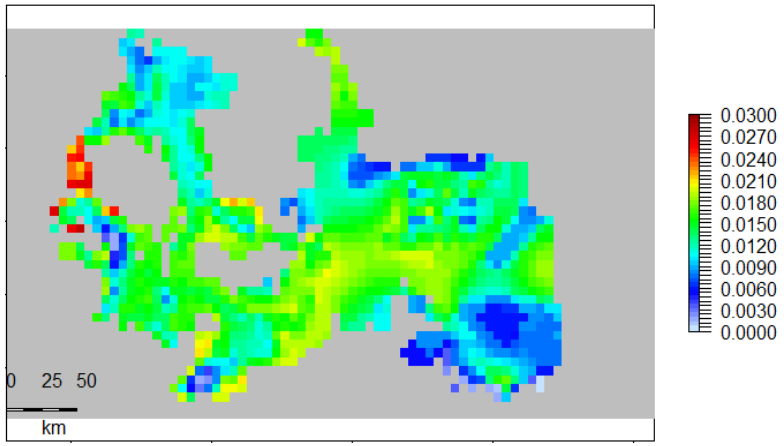


Preliminary distributions

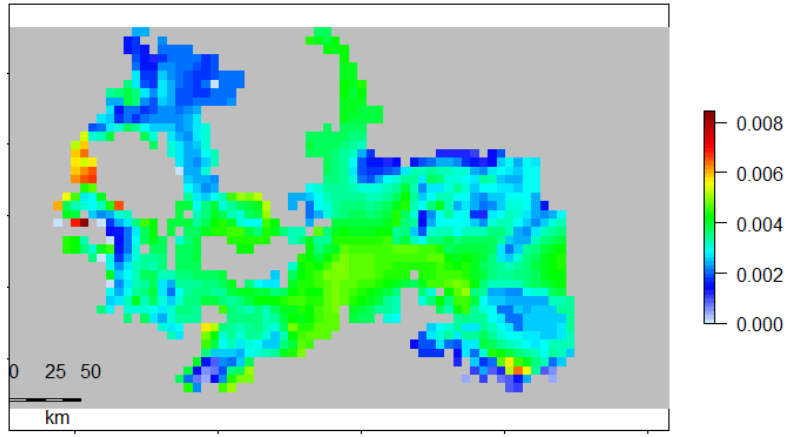
Cod > 35 cm

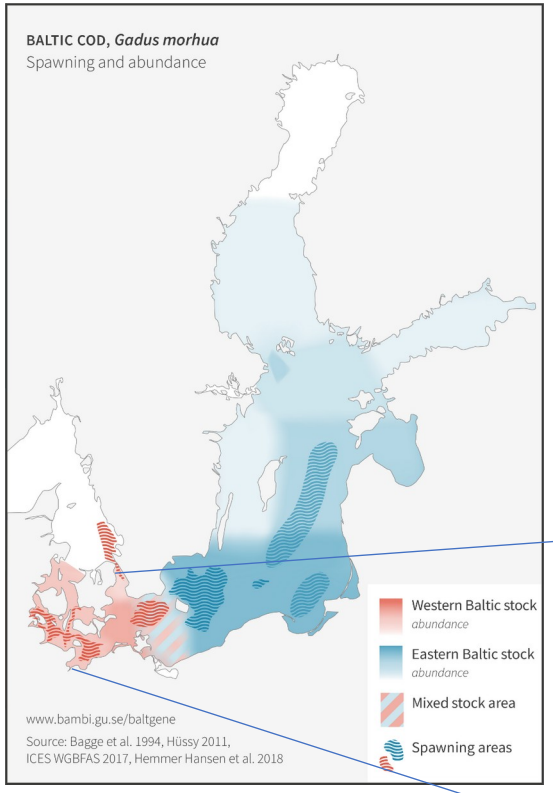


Ecospace predicted distribution
October 1994

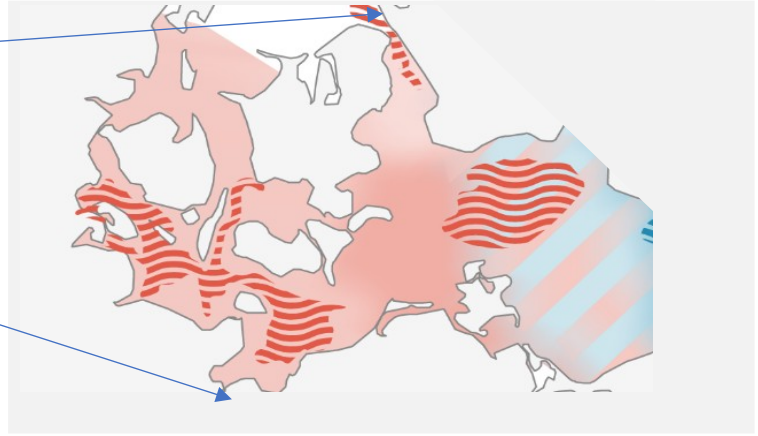


Ecospace predicted catches
October 2023



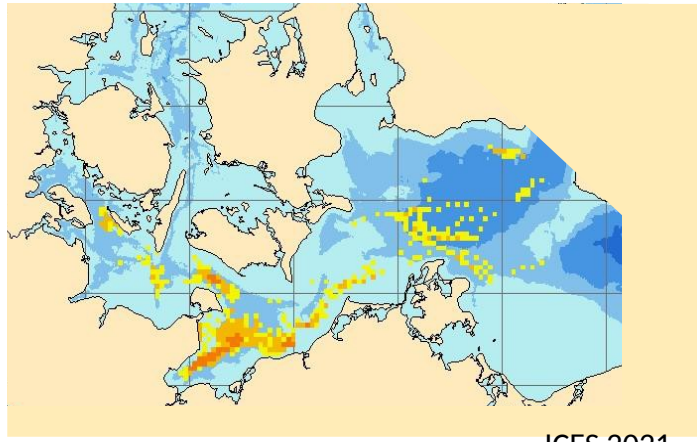


Western baltic cod spawning and abundance

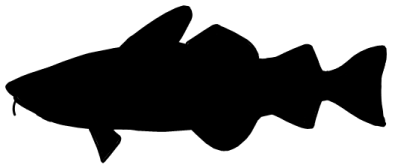


Center for Marine Evolutionary Biology -
University of Gothenburg

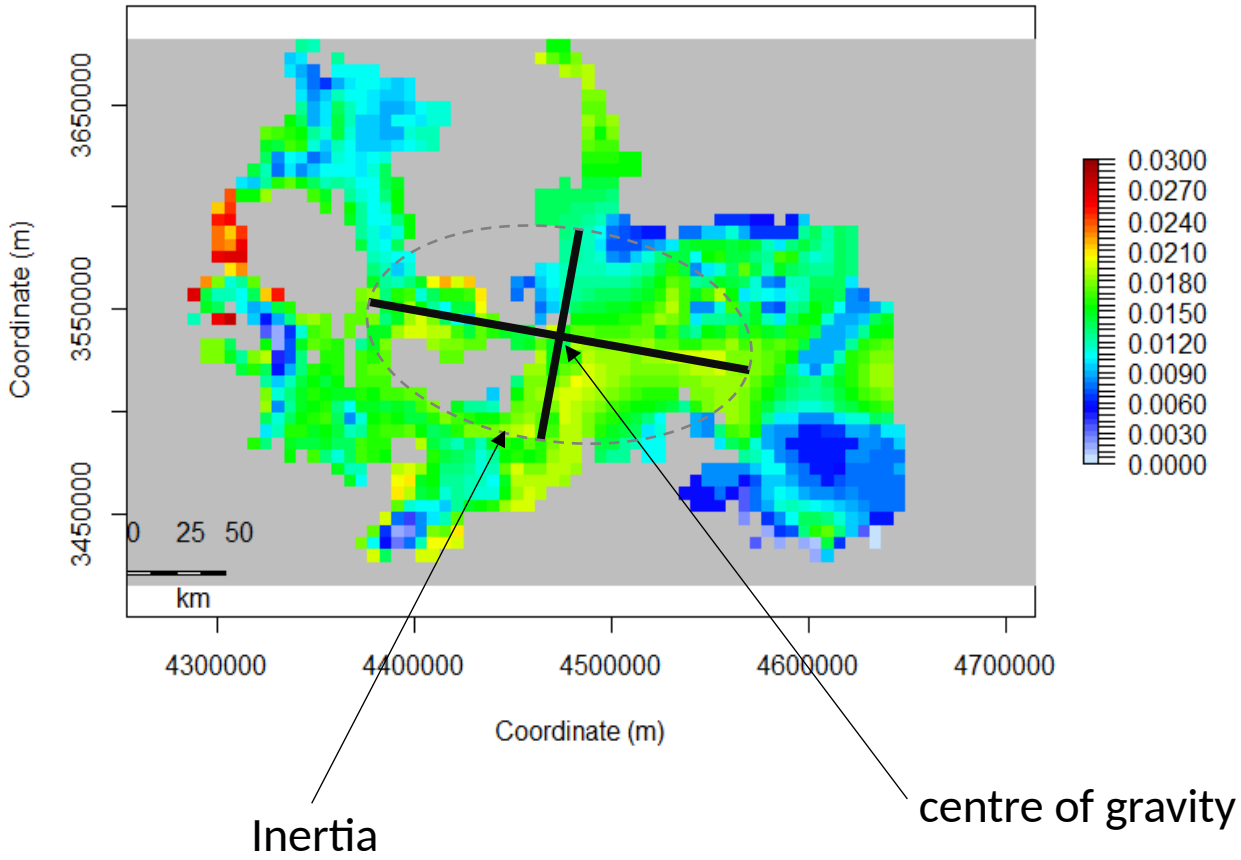
Danish trawl cod directed Fishery 2020



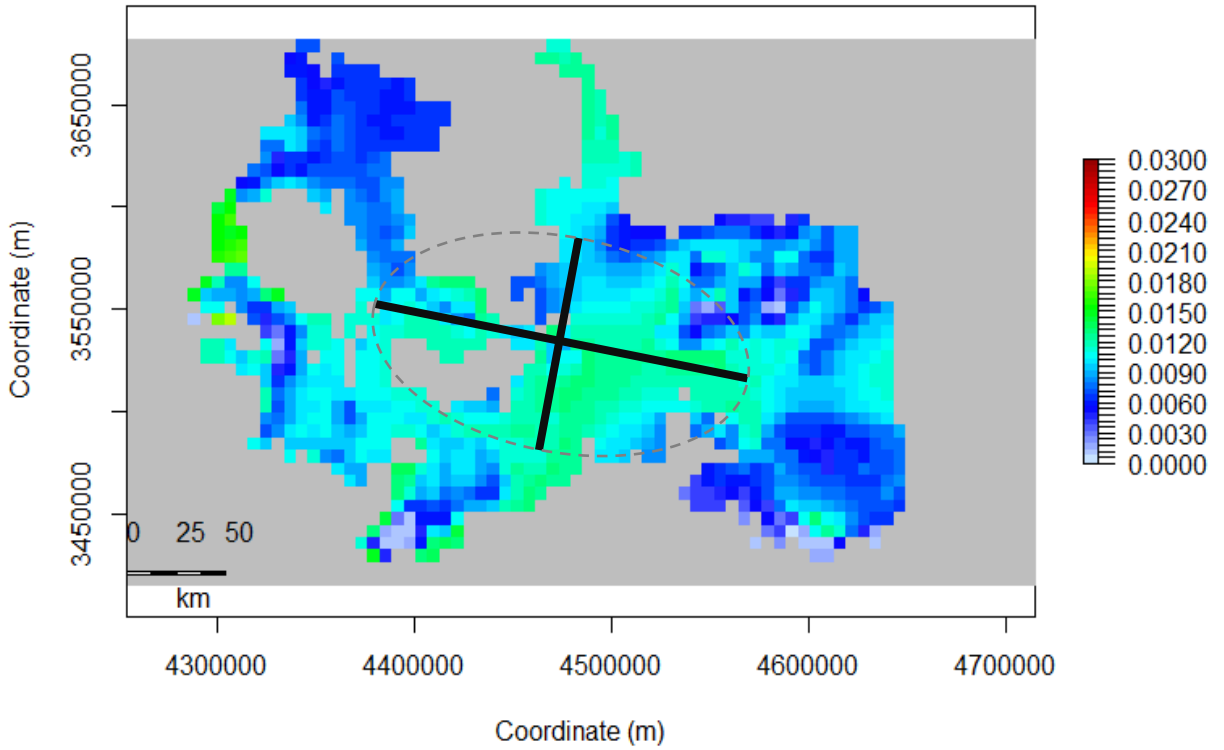
ICES 2021



Relative biomass cod > 35 **1994**



Relative biomass cod > 35 **2023**



Summarising



We would love it, but we don’t have the answers to our questions yet, and we keep working!

Thank you

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